

EE-482: Network Security and Cryptography

LECTURE NO 20: *Firewalls*

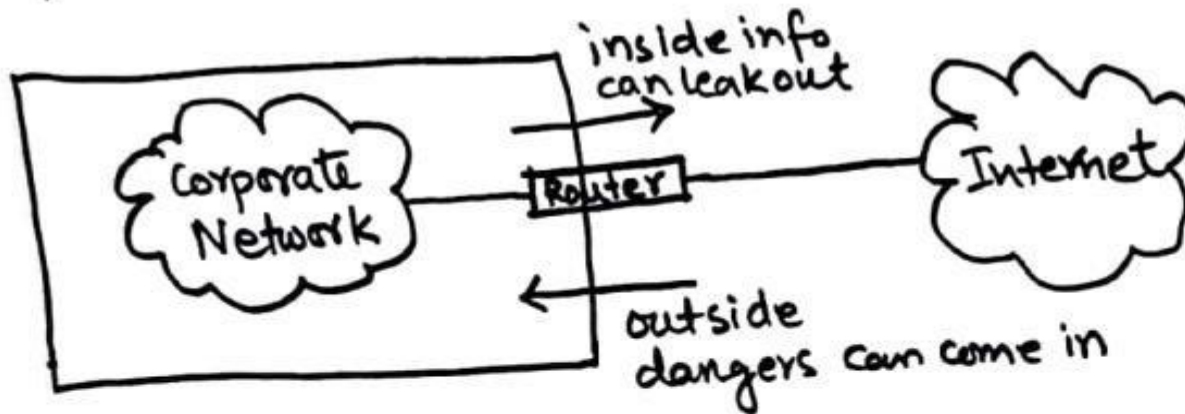
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22. Firewalls (1/19)

Attacks:

- ↳ Most corporations have valuable & confidential data in their networks. It can be leaked to competitors.
- ↳ Danger of outside elements (virus; worms) entering a corporate network to create havoc/disaster.



- ↳ Need mechanism to ensure inside information remains inside and prevent outsiders from entering a network.

22. Firewalls (2/19)

Firewall:

- ↳ A single-choke point that keeps unauthorized users out of protected network and provides protection from various outsider attacks.
- ↳ Use of single chokepoint simplifies security management because security capabilities are consolidated/integrated on a single system point.
- ↳ Firewall provides a location for monitoring security
- ↳ Firewall is a specialized version of router: Basic routing functions and configured to perform firewall functionality

22. Firewalls (3/19)

Firewall characteristics:

- 1) All traffic from inside to outside, and vice versa must pass through the firewall. This is achieved by physically blocking all access to the local network except via firewall.
- 2) Only authorized traffic, as defined by local security policy, will be allowed to pass.
- 3) Firewall itself is immune to penetration. This implies use of a trusted system with secure operating system.

22. Firewalls (4/19)

Access Control:

4 general techniques

- 1) Service Control: Determines types of Internet services that can be accessed.
- 2) Direction Control: Determines direction in which particular service requests may be initiated and allowed to flow through firewall.
- 3) User Control: Control access to a service to which user is attempting to access it. Typically applied to users inside the firewall perimeter.
- 4) Behaviour Control: Controls how particular services are used.

Example: Firewall may filter e-mail to eliminate spam.

22. Firewalls (5/19)

Types of Firewalls:

3 main types

- (a) Packet Filters
- (b) Application-level gateways
- (c) Circuit-level gateways

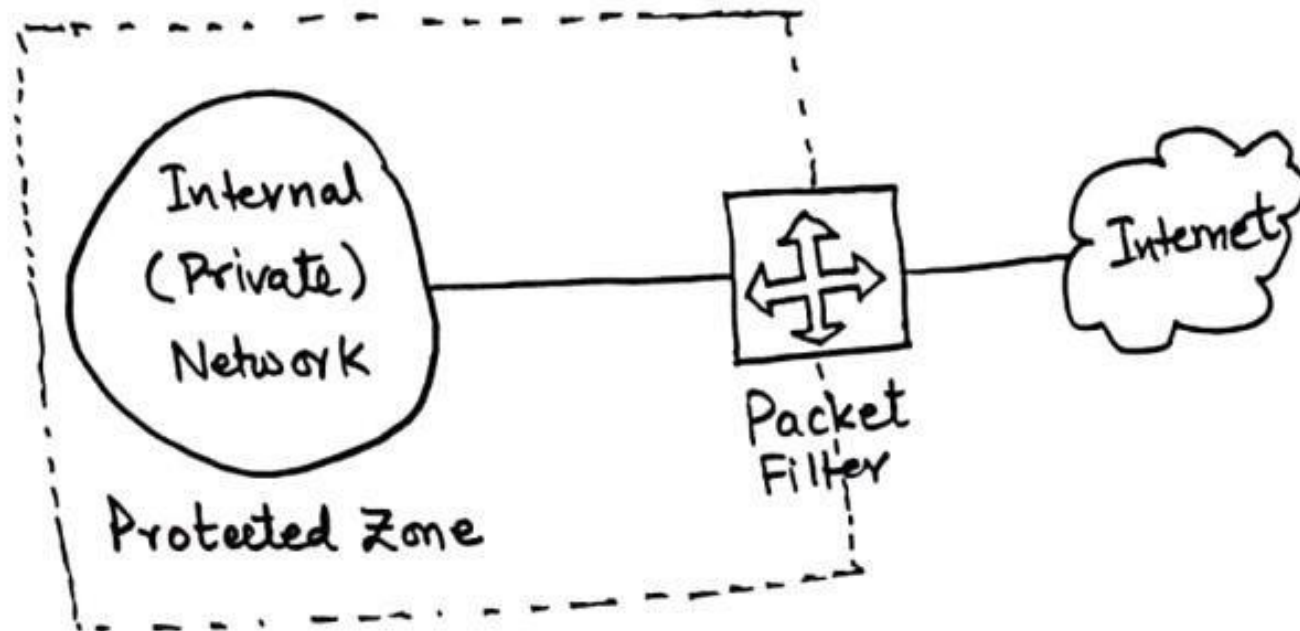
(a) Packet Filters

↳ A packet filter applies a set of rules to each packet & based on the outcome decides to forward or discard the packet.

↳ Also called screening router or screening filter.

22. Firewalls (6/19)

- ↳ Implementation involves a router, configured to filter packets going in either direction.
- ↳ Filtering rules are based on a number of fields in the IP and TCP/UDP headers e.g.: source & destination IP addresses.



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↳ Packet filter performs 3 main functions

- (1) Receive each packet as it arrives
- (2) Pass packet through a set of rules based on contents of IP and transport header fields of the packet. Based on outcome of matching with rules, decide whether to accept or discard packet.
- (3) If there is no match with any rule, take default action
can be:- discard all packets or
- accept all packets

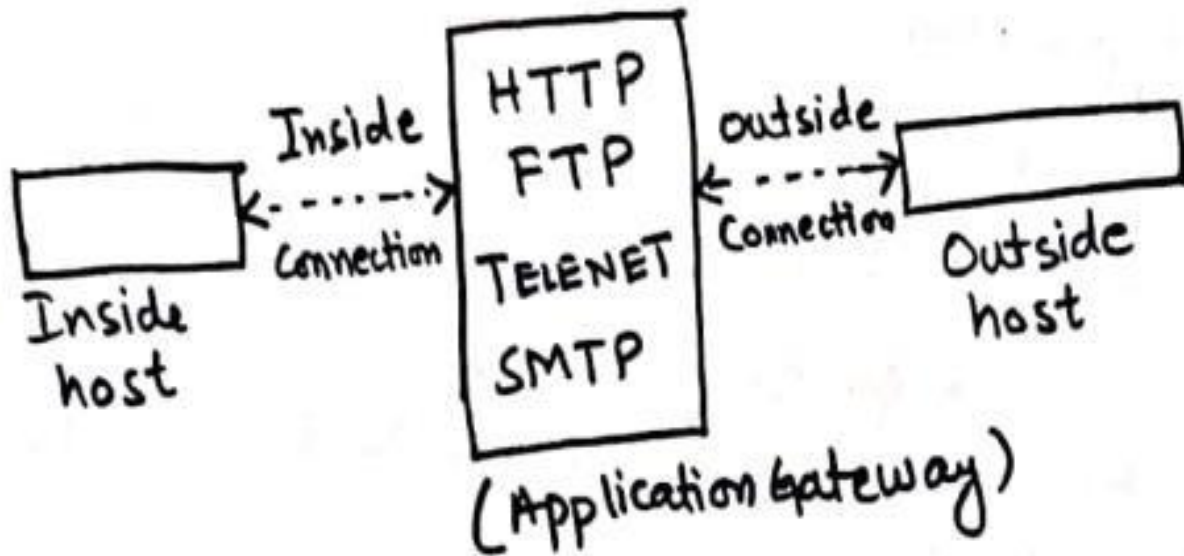
Advantages: → Packet filter is simple
→ Fast operating speeds.

Disadvantages: → difficulties in setting up packet filter rules
→ lack of support for authentication

22. Firewalls (8/19)

(b) Application-Level Gateways:

- ↳ Also called a proxy server.
- ↳ Acts like a proxy (deputy) and decides about the flow of application level traffic.



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Application gateways work as follows:

- 1) User contacts the application gateway using TCP/IP application such as HTTP or TELNET.
- 2) → Gate asks the user for the name of the remote host to be accessed.
→ Also asks for user id and password required to access services of the application gateway.
- 3) User provides this information to gateway.
- 4) Gateway contacts the application on the remote host and relays TCP segments containing application data to the remote host.

Advantages: → more secure than packet filters.
→ easy to log/record & audit all incoming traffic at the application level.

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Disadvantages:

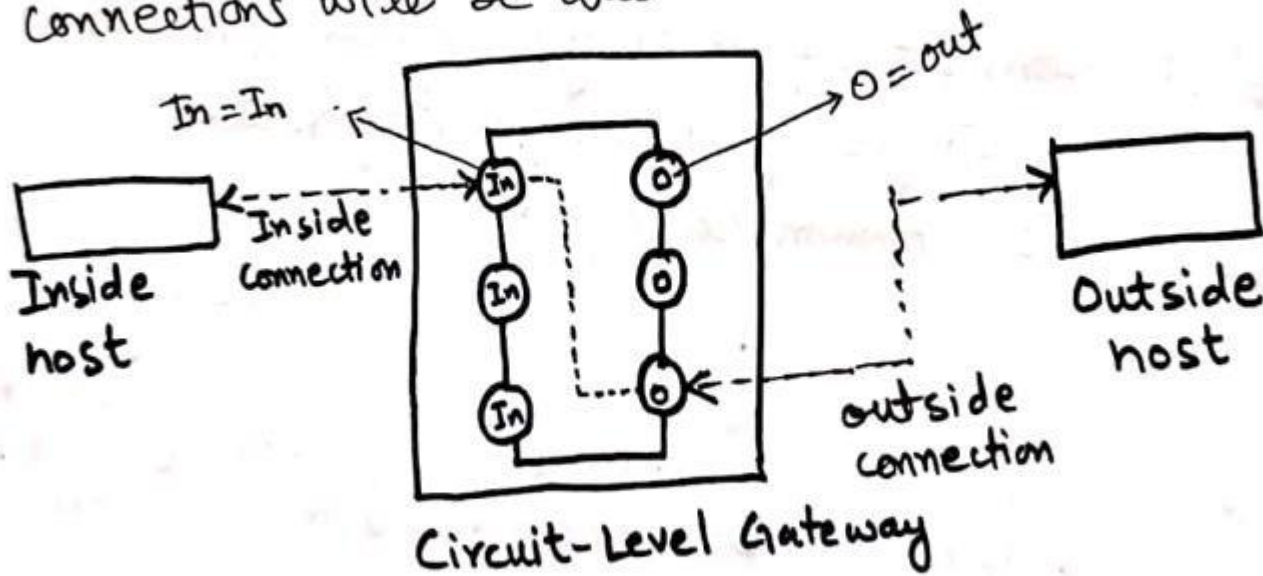
- May need different proxy server for each service.
- Requires modification of client.

(C) Circuit-Level Gateway:

- ↳ Doesn't permit an end-to-end TCP connection; rather gateway sets up two TCP connections, one between itself & TCP user on inner host and second between itself and TCP user on outer host.
- ↳ Once the two connections are established, the gateway relays TCP segments without examining the contents.

22. Firewalls (11/19)

↳ Security function consists of determining which connections will be allowed



↳ Situation in which system administrator trusts internal users

↳ Processing overhead of examining incoming application data but no overhead on outgoing data.

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Bastion Host:

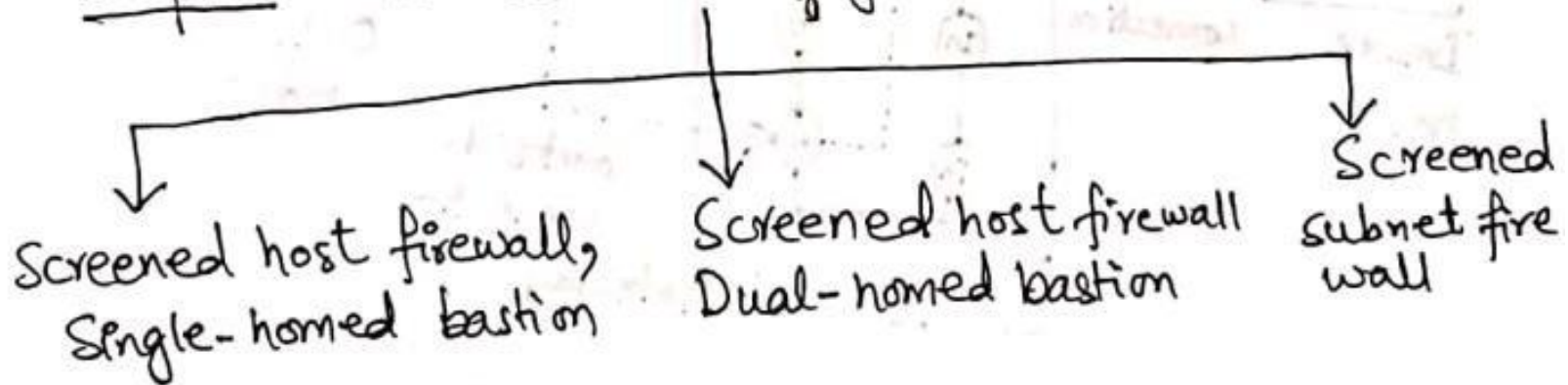
- ↳ A system identified by the firewall administrator as a critical strong point in the network's security.
- ↳ A bastion host serves as a platform for an application-level or circuit-level gateway.
- ↳ Bastion host hardware platform executes a secure version of its operating system making it a trusted system.
- ↳ Only essential services are installed on the bastion host.

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Firewall Configurations

Simple: Consists of a single system such as a single packet-filtering router

Complex: 3 common configurations

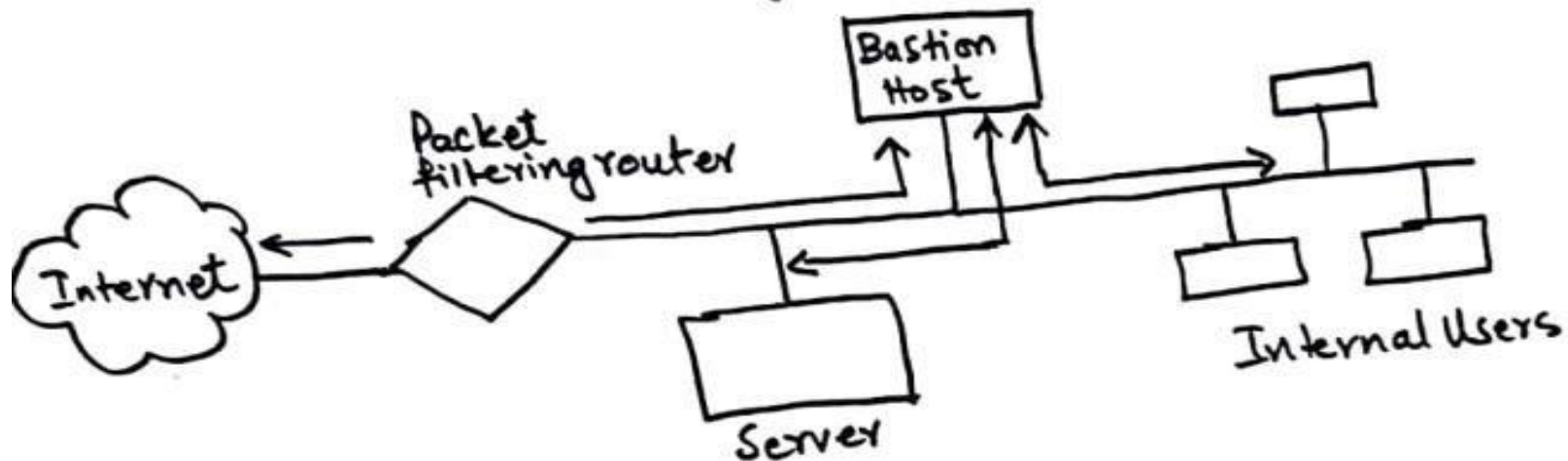


(a) Screened host firewall, Single-homed bastion:

- ↳ Consists of two parts
 - A packet filtering router
 - Bastion host (application gateway).

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- ↳ Packet filter ensures that the incoming traffic is allowed only if it is destined for the application gateway by examining the destination address of every incoming IP packet.
- ↳ Also ensures that outgoing traffic is allowed only if it is originating from application gateway. Done by examining source address.
- ↳ The application gateway / Bastion host performs authentication & proxy functions.



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- ↳ This configuration increases security of network by performing checks at both packet & application level
- ↳ Gives more flexibility to network administrators to define more security policies.

Disadvantage:

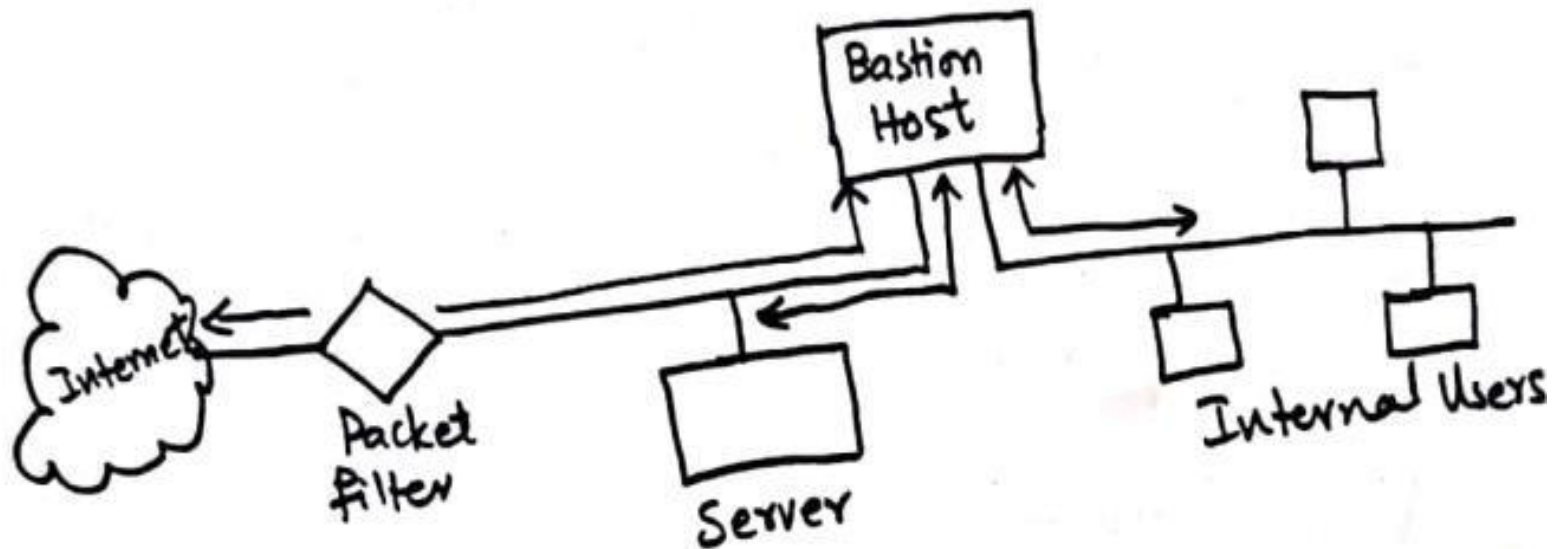
Internal users are connected to bastion host as well as packet filter. If security of packet filter is compromised; whole internal network is exposed to attacker.

Solution: Dual homed Bastion configuration.

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(b) Screened host firewall, Dual-homed Bastion

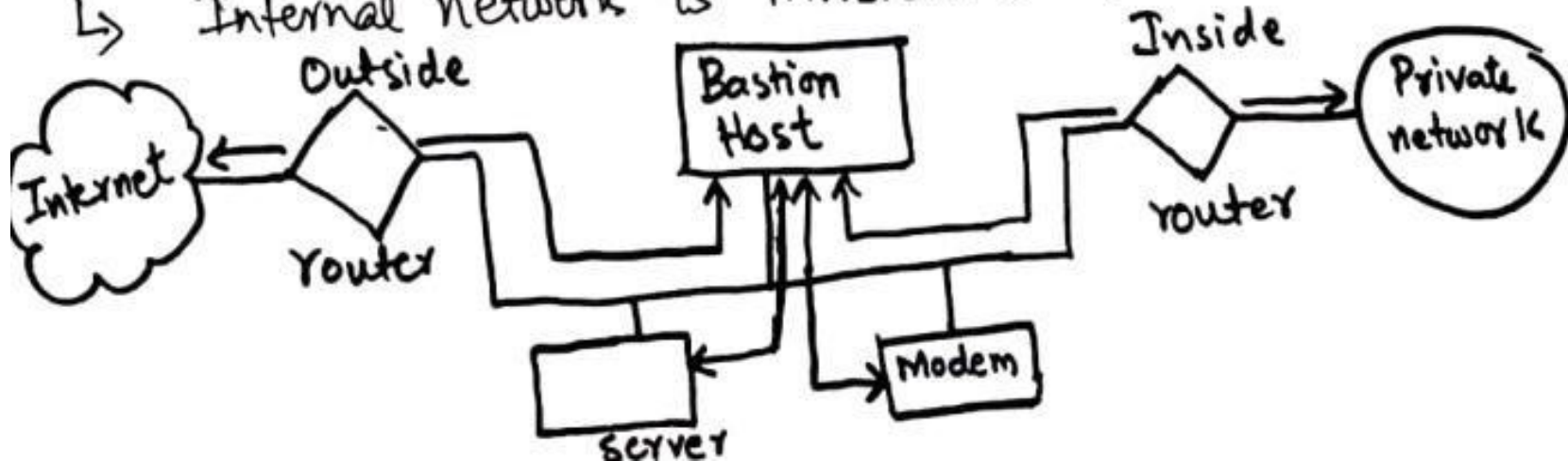
- ↳ Improvement over single-homed Bastion configuration
- ↳ Direct connections between internal hosts & packet filters are avoided
- ↳ Packet filter connects only to application gateway, which has separate connection with internal hosts.



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(C) Screened subnet Firewall:

- ↳ offers highest security.
- ↳ Two packet filters are used, one b/w Internet & Bastion host / application gateway and one between application gateway and internal network.
- ↳ Three levels of defence
- ↳ Attacker doesn't ^{know} about internal network.
- ↳ Internal network is invisible to the Internet



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Limitations of Firewalls:

(1) Insider's Intrusion:

↳ Firewall does not protect against internal threats such as a malicious employee willing to cooperate with an external attacker.

(2) Virus Attacks:

↳ Impossible to scan all incoming files, emails messages for viruses.

↳ Separate virus detection & removal mechanism is required.

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(3) Direct Internet Traffic:

↳ Firewall is effective if it is the only entry-exit point of an internal network.

↳ If firewall is one of the entry points, user can bypass firewall & exchange information via other entry-exit points.

References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6th Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4th Edition, Pearson.*